Derivatives and Fixed Income
PGE M1 Fall 2026

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Session 11: Greeks

BUSINESS SCHOOL
SKEMA

Housekeeping



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Thank you for being part of this course!

Keep in touch!

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Pay attention to the FMI program!

Exam



Midterm

- Covers material from sessions 1-8 + Black-Scholes (Session 10)

Introduction to Option Greeks

Returning to the Black-Scholes framework, the European call on a non-dividend paying stock is given by

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

the European put is given by

$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$$

with

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

Introduction to Option Greeks



We can express the value of the call and put as a function of the inputs,

$$c(S, K, r, \sigma, T)$$

and

$$p(S, K, r, \sigma, T)$$

The “Greeks” allow us to measure the sensitivity of option prices to these inputs.

Delta

The Delta of a (long) call option is defined as

$$\Delta \doteq \frac{\partial c}{\partial S}$$

defined as the rate of change in the option price with respect to the value of the underlying asset. When applied to the price of a European call, is given by

$$\frac{\partial c}{\partial S} = N(d_1) + SN'(d_1)\frac{\partial d_1}{\partial S} - Ke^{-rT}N'(d_2)\frac{\partial d_2}{\partial S}$$

After some algebra and cancelling,

$$\Delta_c = N(d_1)$$

Delta

The Delta of a European call on a non-dividend paying stock is given by

$$\Delta_c = N(d_1)$$

When $S = K$ (the option is ATM), $\Delta_c \approx 0.5$. This means that if the stock price goes up by \$1, the value of the option goes up by about \$0.5.

- For deep OTM options, $\Delta_c \to 0$
- For deep ITM options, $\Delta_c \to 1$

Delta

The Delta of a European put on a non-dividend paying stock is given by

$$\Delta_p = N(d_1) - 1$$

which is bounded between -1 and 0. As the value of the stock rises, the value of the put decreases (for all other inputs held fixed).

Recall: Straddle



A straddle is an options trading strategy that is a combination of a long call and a long put at the same strike price K and expiration date T. Assume you enter a straddle on Apple stock at $K = 350$ with expiry in 1 year. Assume $r = 5\%$, and $c = 14.29, p = 23.81$. For simplicity, assume Apple does not pay dividends.

a. Find the combined payoff of this strategy using tables
b. Plot the payoff diagram
c. Plot the profit diagram

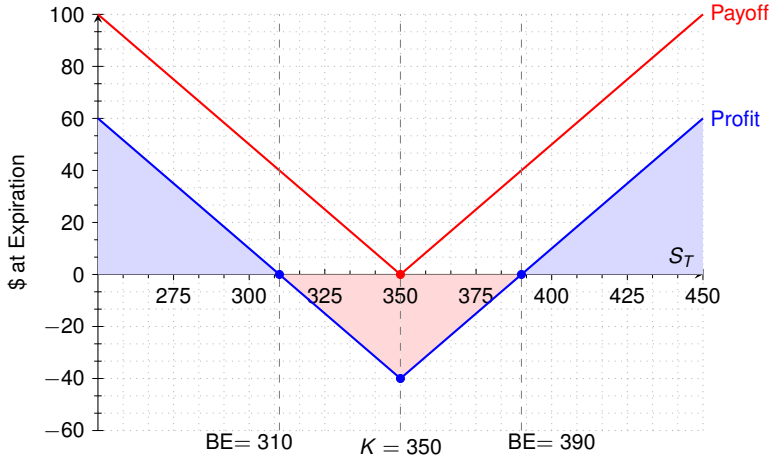
Straddle

a. Find the combined payoff of this strategy using tables.

	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Call	-14.29	0	$S_T - 350$
Long Put	-23.81	$350 - S_T$	0
Total Payoff	-38.10	$350 - S_T$	$S_T - 350$

Straddle

c. Plot the profit diagram. Assume you paid a combined premium of \$38.10 for the call and put ($38.10 \times 1.05 = 40$).



Delta of a Long Straddle



A straddle is commonly written on ATM call and put options. The Delta of this strategy is given by:

Call delta: $N(d_1)$
Put delta: $N(d_1) - 1$

$$\Delta = N(d_1) + [N(d_1) - 1] = 2N(d_1) - 1$$

At at-the-money strikes, $N(d_1) \approx .5$, so $2N(d_1) - 1 \approx 0$. We say that this strategy is delta-neutral.

Delta Hedging (**)

Recall

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

As time passes,

$$d_1 = \frac{\ln(S_t/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}}$$

As the stock price changes and the time to maturity shortens. The value of Δ changes, so a trader maintaining a certain delta position must rebalance his/her portfolio.

Delta Hedging (**)

Example. Suppose a trader sells 50 ATM calls at $K = 100$ with a 1-month expiry. The trader simultaneously buys futures contracts to maintain a delta neutral position.

- **At $t = 0$.** Let $S_0 = 100$. Assume the delta of each call = 0.50. Then, the option delta exposure is

$$50 \text{ contracts} \times 0.50 = 25$$

Since the trader is short call, the trader is -25 delta.

To offset this exposure, the trader can long 25 futures contracts,

$$-25 + 25 = 0 \quad \text{Net Delta}$$

so the portfolio is delta-neutral.

Delta Hedging (**)

Example. Suppose a trader sells 50 ATM calls at $K = 100$ with a 1-month expiry. The trader simultaneously buys futures contracts to maintain a delta neutral position.

- **At $t = 1$.** Let $S_1 = 105$. Assume the new delta of each call = 0.60. Then, the option delta exposure is

$$50 \text{ contracts} \times 0.60 = 30$$

The trader is currently under-hedged with a net delta: $-30 + 25 = -5$.

To be delta-neutral, the trader purchases 5 more futures contracts such that $-30 + 30 = 0$.

Vega

The Vega of a (long) call option is defined as

$$\nu \doteq \frac{\partial c}{\partial \sigma}$$

defined as the rate of change in the option price with respect to the change in the volatility of the underlying asset.

For both (long) European calls and puts on a non-dividend paying underlying, the vega is given by

$$\nu = S_0 \sqrt{\frac{T}{2\pi}} e^{-(d_1)^2/2}$$

Option vega is always positive.

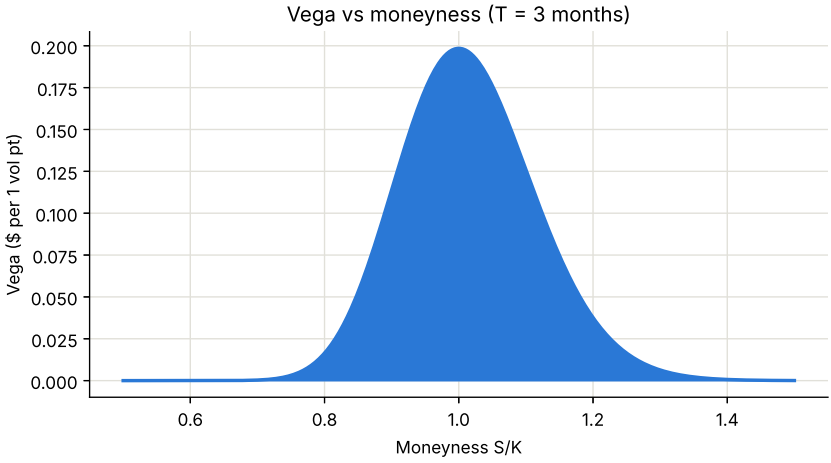
Vega

Even though BS assumes constant σ, we treat σ_t in real-life.
As uncertainty increases, the probability that the stock price take extreme values also increases ($\rightarrow$good for exercising an option).

As such,

- ν is highest for ATM options: not clear whether we will end up ITM or OTM, so an increase in volatility has highest impact on probability of exercise.

Vega (**)



Vega of a call option, calculated for $K = 100$, $T = 0.25$, $r = 2\%$, $\sigma = 20\%$.

Vega of a Straddle

The straddle is a delta-neutral strategy (it has no exposure to the direction of price movements). The vega of the straddle is

$$\nu_{\text{straddle}} = 2S_0\sqrt{\frac{T}{2\pi}}e^{-(d_1)^2/2} \approx 0.8S_0\sqrt{T}$$

at ATM strikes. The long straddle is vega positive, as it bets on volatility.

Conversely, the short straddle is vega negative.

Theta

The Theta of a (long) call option is defined as

$$\Theta \doteq \frac{\partial c}{\partial t}$$

defined as the rate of change in the option price with respect to the passage of time, all else equal (including the stock price).

Theta is usually refered to as the time decay of an option, and is generally negative for long positions and positive for short positions (for both calls and puts).

Theta (**)

For a European call on a non-dividend paying stock,

$$\Theta_c = -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$$

For a European put,

$$\Theta_p = -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT}N(-d_2)$$

where

$$N'(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$$

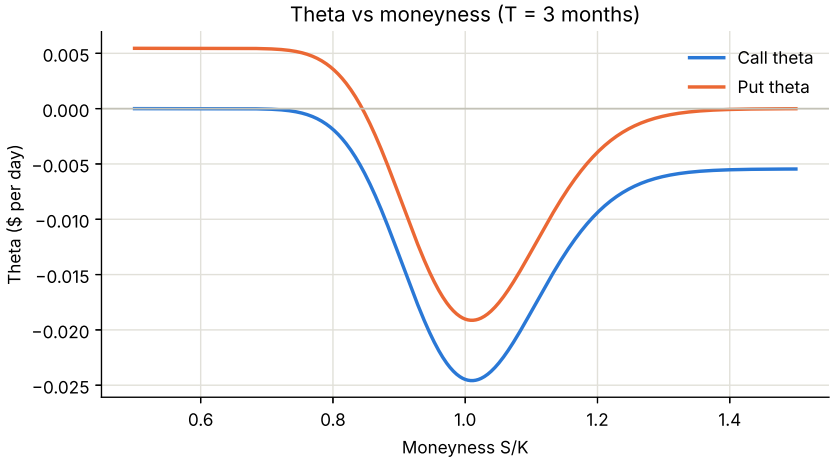
Theta (**)

Unlike the other Greeks, the passage of time is not something you can hedge against. However, it is still useful because to an approximation, the change in the value of your portfolio ΔV :

$$\Delta V \approx \Theta \times \Delta t + \Delta \times \Delta S + \frac{1}{2}\Gamma(\Delta S)^2$$

so knowing how your portfolio changes every day due to Θ helps you pin your exposure to Γ.

Theta (**)



Theta calculated for $K = 100$, $T = 0.25$, $r = 2\%$, $\sigma = 20\%$.

Gamma

The Gamma of a (long) call option is defined as

$$\Gamma \doteq \frac{\partial^2 c}{\partial S^2}$$

It is the second derivative of the option price with respect to underlying asset price, the rate of change of delta. Gamma on a long position is always positive.

When Γ is large (in absolute terms), delta is very sensitive to the price of the underlying asset, and a delta-neutral trader must frequently rebalance his/her portfolio.

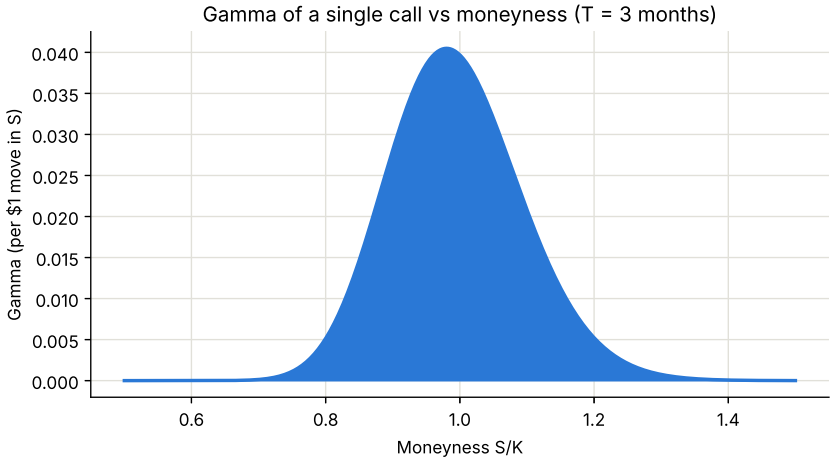
Gamma (**)

For a European call or put on a non-dividend paying stock,

$$\Gamma = \frac{N'(d_1)}{S_0 \sigma \sqrt{T}}$$

Recall, the Delta of a deep ITM option is close to 1. However, the Gamma of a deep ITM option is close to 0 (because Delta is close to 1 and it hardly changes). Gamma, like theta and vega, is highest for ATM options.

Gamma (**)



Gamma of a call option calculated for $K = 100$, $T = 0.25$, $r = 2\%$, $\sigma = 20\%$.

Rho

The rho of a (long) call option is defined as

$$\rho \doteq \frac{\partial c}{\partial r}$$

defined as the rate of change in the option price with respect to a change in the interest rate.

The rho of a long call is positive while the rho of a long put is negative.

Rho (**)

For a European call option on a non-dividend paying stock,

$$\rho_c = KTe^{-rT}N(d_2)$$

while for a European put on a non-dividend paying,

$$\rho_p = -KTe^{-rT}N(-d_2)$$

Using $N(-d_2) = 1 - N(d_2)$,

$$\rho_c - \rho_p = KTe^{-rT} > 0$$

High interest rates decrease the present value of the K that long call pays while the long put receives.

Summary of Greeks

Memorize!

	Δ	ν	Θ	Γ	ρ
Long Call	$+$	$+$	$-$	$+$	$+$
Short Call	$-$	$-$	$+$	$-$	$-$
Long Put	$-$	$+$	$-$	$+$	$-$
Short Put	$+$	$-$	$+$	$-$	$+$